



Python in Research, Part 5

Visualisation

The power of Mayavi/VTK is such that just a few lines of code can create beautiful imagery, which can help you get an insight into scientific data.

Scientific visualisation is the representation of data graphically so that you can gain an understanding and insight into the data. It is useful in virtually every domain of science. For instance, 'bonds in chemistry' was among the first subjects that 3D visualisation helped in gaining an insight into.

The simplest molecule would be a hydrogen molecule ion. You can approximate its orbital functions as a linear combination of atomic orbitals. In this article, the intention is to take a very simple subject and explore the various ways of visualisation.

2D plots

Let the two atoms be at $\pm a$ on the x-axis. The simplest two orbitals will be the bonding and anti-bonding 1s-orbitals. Ignoring the normalisation factors, you can plot the electron density on the x-axis as follows:

```
import scipy as np
a = 0.7 # half the distance between atoms
a0 = 0.53 # Bohr radius in Angstrom
x = np.mgrid[-3.3:3.3]
sa = np.exp(-(x-a)**2/a0)
sb = np.exp(-(x+a)**2/a0)
import matplotlib.pyplot as plt
plt.subplot(2,1,1)
plt.plot(x,sa,'-',label='Bond')
plt.plot(x,sa,'-',label='A')
plt.plot(x,sb,'-',label='B')
plt.legend()
plt.subplot(2,1,2)
```

```
plt.plot(x,np.absolute(sa-sb),'-',label='Anti-bond')
plt.plot(x,sa,'-',label='A')
plt.plot(x,sb,'-',label='B')
plt.legend()
plt.show()
```

You have used *mgrid* instead of *arange* to create an array. The syntax has a hack that if the third parameter is complex, it is interpreted as the number of points rather than the step size.

You have used the *Matplotlib* and it gives a static 2D image, as you can see in Figure 1. If you want to explore the function in other directions, you will have to create a separate plot for each one.

3D visualisation and Mayavi

What if you wished to explore the electron density in a plane? Mayavi comes to your aid. It is a project initiated by Prabhu Ramachandran, and currently enhanced and supported by www.enthought.com. Mayavi offers very easy-to-use Python and Scipy bindings to the visualisation toolkit, VTK, www.vtk.org. You will need to install the Mayavi package. It is available in the Fedora and Ubuntu repositories.

You can plot the function values of the bonding orbital in the xy plane as follows:

```
import scipy as np
x,y = np.mgrid[-3.3:3.3, -3.3:3.3]
a = 0.7
a0 = 0.53
r1=np.sqrt((x-a)**2 + y**2)
r2= np.sqrt((x+a)**2 + y**2)
values = np.exp(-r1/a0) + np.exp(-r2/a0)
```